

ITA0608 - MACHINE LEARNING LAB

PROGRAMS

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Experiment 5: K-NN Algorithm

Aim

To implement the K-Nearest Neighbours (K-NN) algorithm in Python for classification using an appropriate dataset.

Procedure

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Normalize the feature values.
5. Create the K-NN classifier.
6. Train the classifier using the training data.
7. Predict the class labels for the test data.
8. Evaluate the model using accuracy and confusion matrix.
9. Display the results.

Algorithm

1. Start.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Scale the input features.
5. Choose the value of K.
6. Calculate the distance between the test point and all training points.
7. Select the K nearest neighbors.
8. Determine the majority class among the neighbors.

9. Assign the majority class to the test instance.
10. Repeat for all test instances.
11. Calculate the model accuracy.
12. Stop.

Code

```
▶ from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_iris()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = KNeighborsClassifier(n_neighbors=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Labels:")
print(y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Output

```
*** Predicted Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]
Accuracy: 1.0
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Result

The K-Nearest Neighbours (K-NN) algorithm was successfully implemented in Python. The classifier was trained and tested using the Iris dataset, achieving high classification accuracy.

